Adding complexity

Namkha advanced tutorial

Centre for Research into Ecological & Environmental Modelling
University of St Andrews

Session 3 objectives

Main goal: Show where the simple workflow has to change when the analysis gets more realistic.

1. Longer survey periods and closure
2. Shortened-survey and multisession alternatives
3. Elevation constraints and reporting masks
4. Gorge-avoiding movement distances
5. Sex-specific model components

Assumption: You have already seen the simple workflow from Session 2.

From simple to advanced

The basic workflow stays the same:

Each extension affects a specific part of the workflow.

01
camera file
detection file
$\Rightarrow$
`capthist`

02
survey area
buffer
covariates
$\Rightarrow$
`mask`

03
(crop for
multisession
model)
$\Rightarrow$
`mask (ms)`

04
D model
λ_0/g_0
model
$\Rightarrow$
`secr.fit`

05
AICc
abundance
density
maps

Extension 1: Long surveys and closure

Extension 1: Long surveys and closure

Problem: the full survey is long enough that the closed-population assumption may be questionable.

Implementation:

- keep the full survey
- create a shortened survey window
- split the full survey into temporal sessions
- compare how results change

Scripts involved:

- `01_make_capthist.R`
- `03_make_ms_masks.R`
- `04a`, `04b`, `04c`
- `05a`, `05b`, `05c`

Full survey: occasion-specific usage

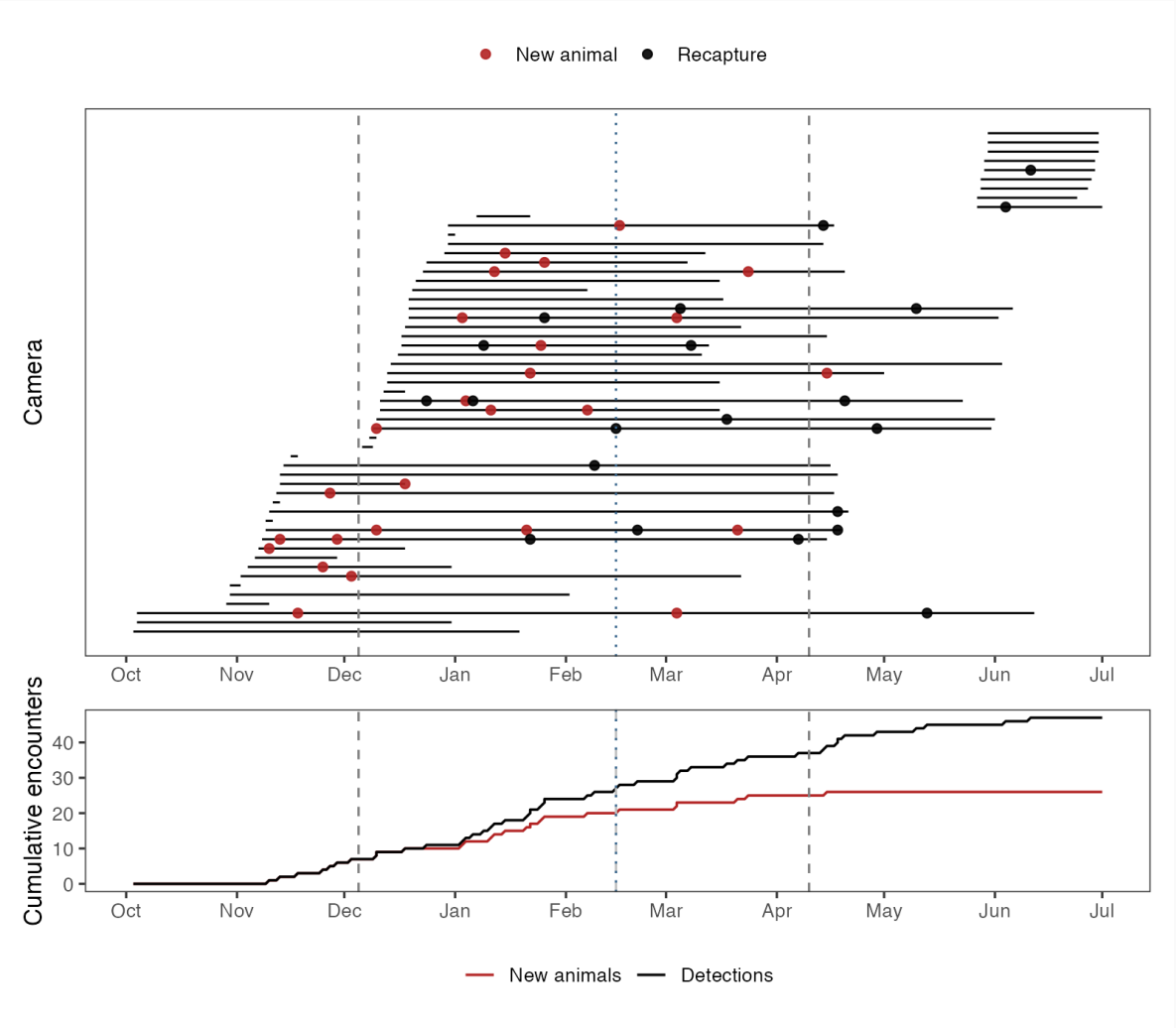
`secr` implements closure checks. These need multiple occasions (no other need for these i.e. for model fitting)

```
full_occ_length <- 14
full_occ_start <- seq.Date(from = survey_start, to = survey_end,
                           by = paste(full_occ_length, "days"))
full_occ_bins <- interval(start = full_occ_start,
                          end = full_occ_end)
```

For each camera and occasion, need to recalculate effort (days active)

```
usage_full <- matrix(0, nrow = nrow(cameras), ncol = length(full_occ_bins))
for (i in seq_len(nrow(cameras))) {
  usage_full[i, ] <- round(as.numeric(
    int_overlaps_numeric(cameras$camera_on[i], full_occ_bins)) / (24 * 60 * 60))
}
usage(full_traps) <- usage_full
```

Full survey: descriptive check



- cameras started at different times
- new animals appear through time
- shortened window is a defensible alternative
- session split is another alternative

Full survey: closure checks

The full survey occasions allow a formal closure check

```
closure.test(full_ch)
```

```
  statistic           p
-1.56677248  0.05858394
```

Suggests moderate concern about closure. Now check closure for the shortened survey

```
short_occasion_start <- floor(as.numeric(short_survey_start - survey_start) /
                              full_occ_length) + 1
short_occasion_end <- floor(as.numeric(short_survey_end - survey_start) /
                            full_occ_length) + 1

closure.test(subset(full_ch, occasions = short_occasion_start:short_occasion_en
```

```
  statistic           p
-0.06098922  0.47568390
```

Shortened survey: workflow change

Key idea: limit survey length by taking a subset of the full survey

Issue: Which part of the full survey to take?

Keep only detections in the shortened window

```
short_capts_raw <- capts_raw |>
  filter(date >= short_survey_start, date <= short_survey_end)
```

Recalculate camera effort for that same window

```
short_traps_df <- cameras |>
  mutate(
    effort_short = pmax(
      0,
      as.numeric(pmin(end_date, short_survey_end) -
                 pmax(start_date, short_survey_start), units = "days")
    )
  )
```

Shortened survey: `capthist`

Create and check the capture history

```
short_capts <- short_capts_raw |>
  mutate(occasion = 1L) |>
  dplyr::select(session, animalID, occasion, trapID, sex)

short_ch <- make.capthist(captures = short_capts, traps = short_traps)
verify(short_ch)
summary(short_ch, terse = TRUE)
```

```
 Occasions Detections     Animals  Detectors
         1         30          21         55
```

Multisession survey: workflow change

Key idea: keep the full survey but split it into temporal sessions.

Detections get a session label:

```
ms_capts <- capts_raw |>
  mutate(
    session = if_else(date < ms_session_end, 1L, 2L),
    occasion = 1L
  ) |> dplyr::select(session, animalID, occasion, trapID, sex)
```

Camera effort is calculated separately for each session:

```
ms_traps_df <- cameras |>
  mutate(
    effort_s1 = pmax(0, as.numeric(pmin(end_date, ms_session_end - days(1)) -
                                     pmax(start_date, survey_start),
                                     units = "days")),
    effort_s2 = pmax(0, as.numeric(pmin(end_date, survey_end) -
                                     pmax(start_date, ms_session_end),
                                     units = "days"))
  )
```

Multisession survey: `capthist`

For multisession `secr`, the traps object is a named list:

```
ms_traps_list <- split(ms_traps_df, f = ms_traps_df$session)
traps_ms <- vector("list", length(ms_traps_list))

for (i in seq_along(ms_traps_list)) {
  traps_ms[[i]] <- read.traps(
    data = ms_traps_list[[i]] |> dplyr::select(trapID, x, y, effort),
    detector = "count",
    trapID = "trapID",
    binary.usage = FALSE
  )
}
names(traps_ms) <- c("1", "2")
```

Then add covariates, etc.

Multisession survey: factor levels

Create and check the multisession capture history:

```
ch_ms <- make.capthist(captures = ms_capts, traps = traps_ms)
verify(ch_ms)
```

Make sure to share factor levels across sessions:

```
ch_ms <- shareFactorLevels(ch_ms, stringsAsFactors = FALSE)
verify(ch_ms)
summary(ch_ms, terse = TRUE)
```

```
  Occasions Detections Animals Detectors
1         1         26      20        55
2         1         21      16        55
```

Extension 2: Elevation constraints and reporting masks

Extension 2: Elevation constraints

Problem: not every point in the trap buffer is plausible snow leopard habitat.

Implementation:

- add elevation to the full mask
- restrict mask points to plausible elevation range
- keep separate masks for fitting and reporting
- save scaling information for covariate effect plots

Add elevation to the mask

Elevation is downloaded and saved if not available already:

```
elev_file <- "namkha_advanced/output/cache/namkha_elevation.tif"

if (file.exists(elev_file)) {
  elev <- rast(elev_file)
} else {
  elev_download <- get_elev_raster(locations = full_region_4326,
                                   z = 10, clip = "locations")
  elev <- rast(elev_download)
  writeRaster(elev, elev_file, overwrite = TRUE)
}
```

Then project and attach it to the mask:

```
elev <- project(elev, y = my_crs)
names(elev) <- "elev"
mask <- addCovariates(mask, elev)
```

Restrict masks by elevation

We use 3000–5500 m as the plausible elevation range.

```
inside_elev_range <- covariates(mask)$elev >= elev_min &
  covariates(mask)$elev <= elev_max
mask_elev <- subset(mask, subset = inside_elev_range)
```

Then create reporting and fitting masks.

```
mask_survey_area_elev <- subset(mask_elev, subset = inside_survey_area_elev)
mask_model <- subset(mask_elev, subset = inside_buffer_elev)
```

Fitting mask versus reporting mask

Fitting mask: `mask_model`

- elevation-constrained
- within the trap buffer
- can include activity centres outside Namkha

Reporting mask: `mask_survey_area_elev`

- elevation-constrained
- inside the Namkha survey boundary
- used for abundance, density, maps, and covariate summaries

Same conceptual split as Session 2, but now with an extra habitat constraint.

Extension 3: Barriers and custom movement distances

Extension 3: Gorge-avoiding movement

Problem: Euclidean distance says animals can move straight through the gorge.

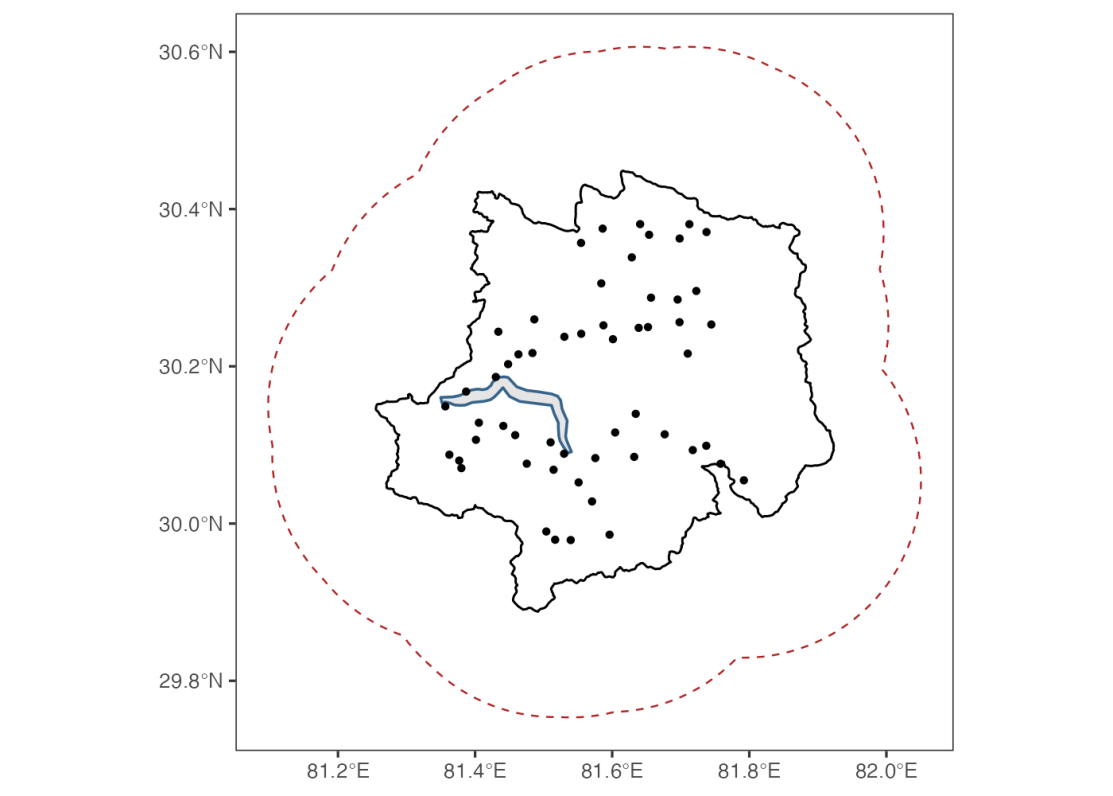
Implementation:

- read the gorge geometry
- make a fine mask for path calculations
- calculate shortest-path distances with `nedist()`
- pass those distances to `secr.fit()` using `details`

Read the gorge

The gorge is a polygon that must be in the same CRS as traps and masks.

```
gorge <- st_read("../namkha_gorge.kml") |>
  st_transform(my_crs) |>
  st_geometry()
```



Fine mask for movement distances

Gorge is narrower than existing mask spacing, so shortest-path calculations will be inaccurate. Use a finer mask to calculate distances around the gorge.

```
mask_fine <- make.mask(
  traps = full_traps,
  buffer = mask_buffer,
  type = "trapbuffer",
  spacing = 200,
  poly = gorge,
  poly.habitat = FALSE,
  keep.poly = FALSE
)
```

Note: Still use the normal model mask for model fitting.

Non-Euclidean distance matrix

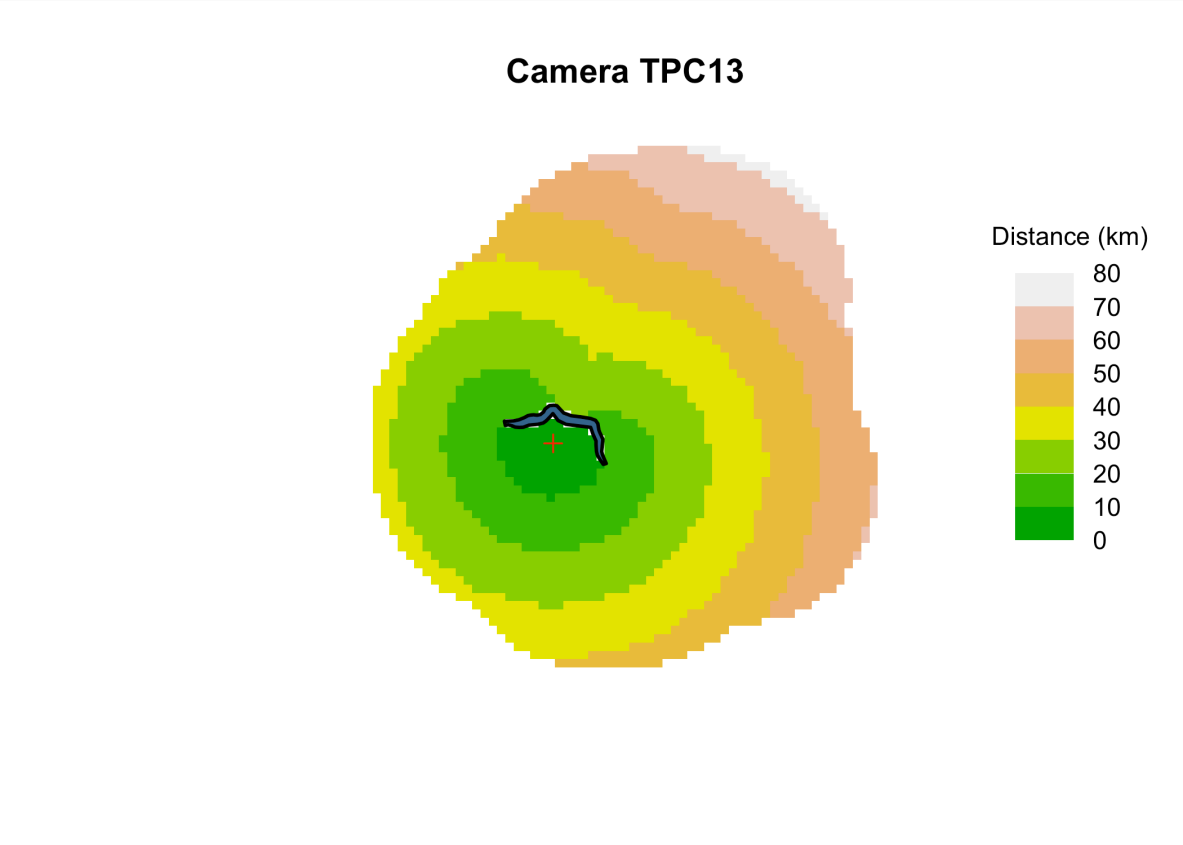
`nedist()` calculates distances from traps to mask points, using the fine mask as the movement network.

```
userd <- nedist(
  full_traps,
  mask,
  mask_fine,
  directions = 16
)
```

Each cropped mask needs its own `userdist` (this is just a subsetting of rows and columns of the full `userdist`)

```
userd_elev <- userd[, inside_elev_range, drop = FALSE]
userd_model <- userd_elev[, inside_buffer_elev, drop = FALSE]
userd_survey_area_elev <- userd_elev[, inside_survey_area_elev, drop = FALSE]
```

Non-Euclidean distance check



- distance from camera ‘TPC13‘ to each mask point
- points across the gorge are farther away than straight-line distance suggests

Use `userdist` in model fitting

This is the key change to `secr.fit()`:

```
f0 <- secr.fit(
  full_ch,
  detectfn = "HHN",
  mask = mask_model,
  model = list(D ~ 1, lambda0 ~ 1, sigma ~ 1),
  details = list(userdist = userd_model),
  verify = FALSE,
  trace = FALSE
)
```

The model still uses `mask_model`; only the movement distances have changed.

Extension 4: Multisession masks and distances

Extension 4: Multisession masks

Problem: `ch_ms` has multiple sessions, so model fitting needs one mask per session.

Implementation:

- build a trap-buffer polygon for each session
- crop the elevation-constrained mask to each session
- create one `userdist` matrix per session
- keep names aligned with `traps_ms`

One mask per session

Create polygons around traps active in each session:

```
mask_per_session_sf <- list()
for (i in seq_along(traps_ms)) {
  mask_per_session_sf[[i]] <- st_as_sf(traps_ms[[i]],
                                       coords = c("x", "y"), crs = my_crs) |>
    st_buffer(mask_buffer) |>
    st_union()
}
names(mask_per_session_sf) <- names(traps_ms)
```

Crop the full elevation mask:

```
mask_ms <- list()
for (i in seq_along(mask_per_session_sf)) {
  mask_in <- lengths(st_intersects(mask_elev_sf,
                                   mask_per_session_sf[[i]])) > 0
  mask_ms[[i]] <- subset(mask_elev, subset = mask_in)
}
names(mask_ms) <- names(traps_ms)
```

One `userdist` per session

Use the same mask indicator to subset the full distance matrix:

```
userd_ms <- list()
for (i in seq_along(mask_per_session_sf)) {
  mask_in <- lengths(st_intersects(mask_elev_sf,
                                   mask_per_session_sf[[i]])) > 0
  userd_ms[[i]] <- userd_elev[, mask_in, drop = FALSE]
}
names(userd_ms) <- names(traps_ms)
```

The names of `mask_ms`, `userd_ms`, and `traps_ms` should all match.

Fit multisession models

The fitting call uses lists for `mask` and `userdist`:

```
ms0 <- secr.fit(
  ch_ms,
  detectfn = "HHN",
  mask = mask_ms,
  model = list(D ~ 1, lambda0 ~ 1, sigma ~ 1),
  details = list(userdist = userd_ms),
  verify = FALSE,
  trace = FALSE
)
```

Multisession analyses also good when sessions are spatial (geographically separate surveys that you want to combine)

Multisession model set

Can share parameters across sessions:

```
ms1 <- secr.fit(ch_ms, detectfn = "HHN", mask = mask_ms,
                model = list(D ~ std_tri, lambda0 ~ 1, sigma ~ 1),
                details = list(userdist = userd_ms), start = ms0)
```

Or include session as a covariate:

```
ms5 <- secr.fit(ch_ms, detectfn = "HHN", mask = mask_ms,
                model = list(D ~ session, lambda0 ~ 1, sigma ~ 1),
                details = list(userdist = userd_ms), start = ms0)
```

Note: How to extrapolate from session-specific covariate models to a wider region is unclear.

Extension 5: Sex-specific models

Extension 5: Sex-specific models

Problem 1: detection or density may differ by sex Problem 2: sex not known for all animals

Two approaches:

1. `hcov = "sex"`: individual covariate models (Problem 1 & 2)
2. `groups = "sex"`: grouped real-parameter models (Problem 1 only)

Important: do not compare the `hcov` and `groups` models with each other (or with other secr models) using AIC.

Sex as an individual covariate

`hcov = "sex"` are finite mixture models that retain animals with unknown sex.

```
f0 <- secr.fit(
  full_ch,
  detectfn = "HHN",
  mask = mask_model,
  model = list(D ~ 1, lambda0 ~ h2, sigma ~ 1),
  hcov = "sex",
  details = list(userdist = userd_model),
  verify = FALSE
)
```

`h2` is the term used by `secr` for the first two levels of the individual covariate.

Sex as groups

Grouped models estimate separate real parameters by sex, but animals with unknown sex must be removed:

```
full_ch_known_sex <- subset(
  full_ch,
  subset = covariates(full_ch)$sex != "Unknown"
)
full_ch_known_sex <- shareFactorLevels(full_ch_known_sex)
```

Then fit grouped models:

```
f0g <- secr.fit(
  full_ch_known_sex,
  detectfn = "HHN",
  mask = mask_model,
  model = list(D ~ 1, lambda0 ~ g, sigma ~ 1),
  groups = "sex",
  details = list(userdist = userd_model)
)
```

Sex-specific model comparison

Compare each sex-specific candidate set internally:

```
AIC(sex_f0, sex_f1, sex_f2, sex_f3, sex_f4, criterion = "AICc")
```

```
       modelname    AICc dAICc AICcwt
sex_f3    sex_f3 261.154 0.000 0.6539
sex_f4    sex_f4 264.144 2.990 0.1466
sex_f0    sex_f0 264.220 3.066 0.1412
sex_f2    sex_f2 267.302 6.148 0.0302
sex_f1    sex_f1 267.454 6.300 0.0280
```

Sex-specific model comparison

Compare each sex-specific candidate set internally:

```
AIC(sex_f0g, sex_f1g, sex_f2g, sex_f3g, sex_f4g, criterion = "AICc")
```

```
                                                        model          detectfn
sex_f3g                 D~1 lambda0~valley_stream + g sigma~1 hazard halfnormal
sex_f4g D~std_tri lambda0~cliff + valley_stream + g sigma~1 hazard halfnormal
sex_f0g                                  D~1 lambda0~g sigma~1 hazard halfnormal
sex_f1g                           D~std_tri lambda0~g sigma~1 hazard halfnormal
sex_f2g                          D~std_elev lambda0~g sigma~1 hazard halfnormal
        npar    logLik     AIC    AICc dAICc AICcwt
sex_f3g    5 -194.3884 398.777 402.306 0.000 0.8880
sex_f4g    7 -193.1579 400.316 407.782 5.476 0.0575
sex_f0g    4 -199.3268 406.654 408.876 6.570 0.0332
sex_f1g    5 -198.7941 407.588 411.118 8.812 0.0108
sex_f2g    5 -198.8246 407.649 411.179 8.873 0.0105
```

```
        modelname    AICc dAICc AICcwt
sex_f3g   sex_f3g 402.306 0.000 0.8880
sex_f4g   sex_f4g 407.782 5.476 0.0575
```

Sex-specific model comparison

Finite mixtures, best model:

```
predict(sex_f3)
```

```
$`session = 1, h2 = Female, valley_stream = 0`
          link     estimate  SE.estimate          lcl           uc
D          log 1.592746e-04 4.377542e-05 9.385261e-05 2.703006e-0
lambda0    log 1.969504e-03 1.354382e-03 5.816625e-04 6.668724e-0
sigma      log 7.657208e+03 1.133651e+03 5.737628e+03 1.021900e+0
pmix     logit 3.761775e-01 1.285911e-01 1.708228e-01 6.383471e-0

$`session = 1, h2 = Male, valley_stream = 0`
          link     estimate  SE.estimate          lcl           uc
D          log 1.592746e-04 4.377542e-05 9.385261e-05 2.703006e-0
lambda0    log 2.581066e-03 8.225013e-04 1.403074e-03 4.748076e-0
sigma      log 7.657208e+03 1.133651e+03 5.737628e+03 1.021900e+0
pmix     logit 6.238225e-01 1.285911e-01 3.616529e-01 8.291772e-0
```

Sex-specific model comparison

Groups, best model:

```
predict(sex_f3g)
```

```
$`session = 1, g = Female, valley_stream = 0`
         link     estimate  SE.estimate          lcl          ucl
D         log 8.181020e-05 2.537140e-05 4.517155e-05 1.481665e-04
lambda0   log 1.114858e-03 6.065400e-04 4.109557e-04 3.024436e-03
sigma     log 8.204458e+03 1.407858e+03 5.875501e+03 1.145658e+04

$`session = 1, g = Male, valley_stream = 0`
         link     estimate  SE.estimate          lcl          ucl
D         log 8.181020e-05 2.537140e-05 4.517155e-05 1.481665e-04
lambda0   log 2.253713e-03 7.589529e-04 1.185533e-03 4.284337e-03
sigma     log 8.204458e+03 1.407858e+03 5.875501e+03 1.145658e+04
```

Extension 6: Advanced reporting

Extension 6: Advanced reporting

Reporting follows the Session 2 pattern:

- choose fitted models
- choose reporting mask
- estimate abundance with `region.N()`
- convert abundance to density
- join with AICc
- make maps and covariate-effect plots

Main change: If reporting over a new mask, replace `userdist` with the matching distance matrix.

Other reporting code works more or less like before (see scripts and vignette for details).

Reporting mask and `userdist`

Choose the reporting mask:

```
extrap_mask <- mask_survey_area_elev
extrap_mask_userd <- userd_survey_area_elev
```

Copy the fitted models and swap the distance matrix:

```
fitted_models_region <- lapply(fitted_models, function(model) {
  model$details$userdist <- extrap_mask_userd
  model
})
```

Now fitted models stored in `fitted_models_region` can be used to extrapolate to the reporting mask (e.g. abundance, maps).

Comparing full, short, multisession surveys

Full survey:

	modelname	AICc	dAICc	estimate	D	CV
1	f3	225.08	0.00	33.32	0.02	25.91
2	f4	226.39	1.31	34.14	0.02	25.73
3	f0	228.76	3.68	33.23	0.02	25.78
4	f2	231.24	6.16	33.71	0.02	25.27
5	f1	231.38	6.30	33.55	0.02	25.29

Short survey:

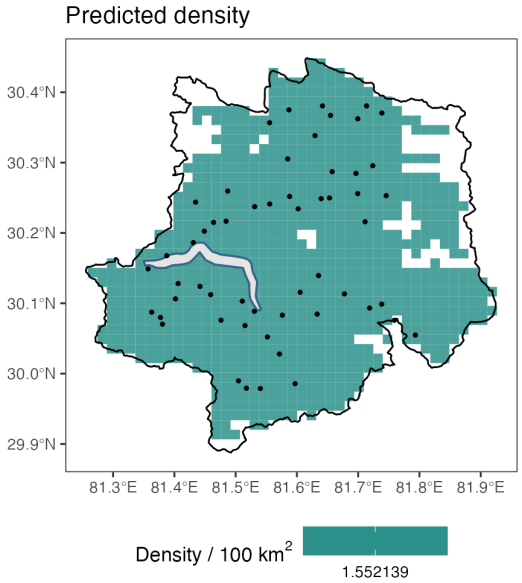
	modelname	AICc	dAICc	estimate	D	CV
1	s3	138.29	0.00	45.19	0.02	36.39
2	s0	141.70	3.41	44.50	0.02	37.16
3	s4	141.78	3.48	48.67	0.02	38.67
4	s1	144.28	5.99	44.71	0.02	37.17
5	s2	144.79	6.49	44.44	0.02	36.13

Multisession survey may return session-specific results (not here).

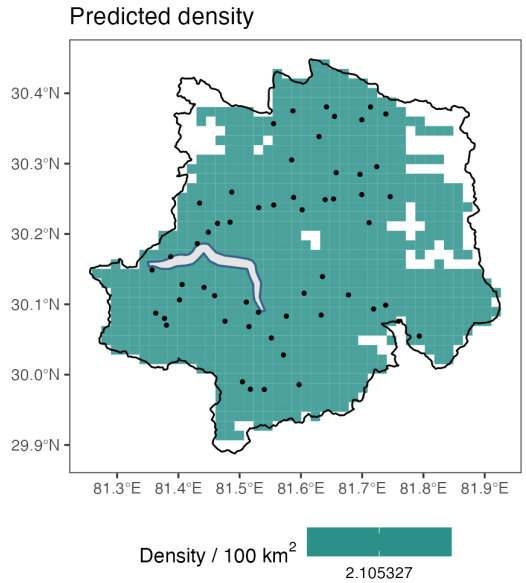
	modelname	AICc	dAICc	estimate	D	CV
1	ms0	221.72	4.20	40.64	0.02	32.88
2	ms1	224.25	6.74	40.69	0.02	32.20
3	ms2	224.25	6.74	40.76	0.02	32.33
4	ms3	217.82	0.31	41.07	0.02	32.78

Predicted density maps

Full survey

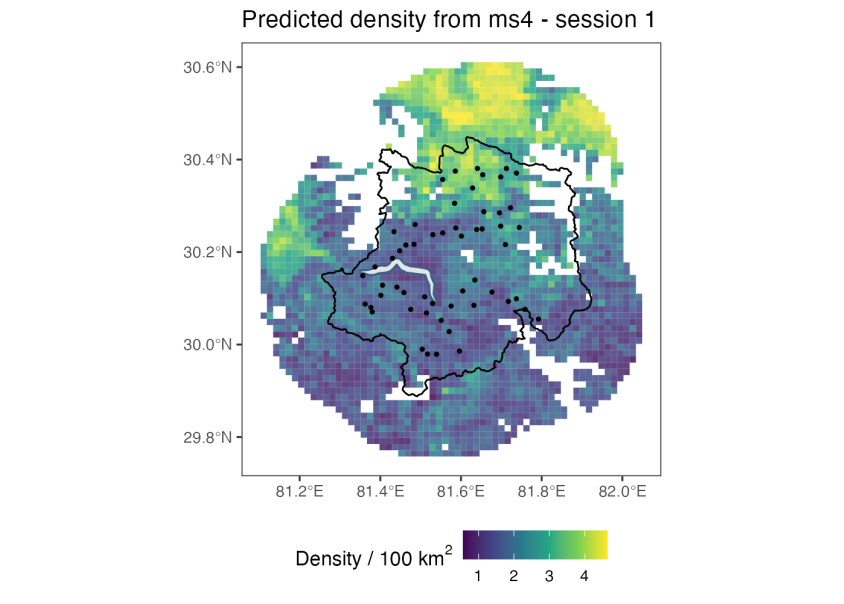


Shortened survey

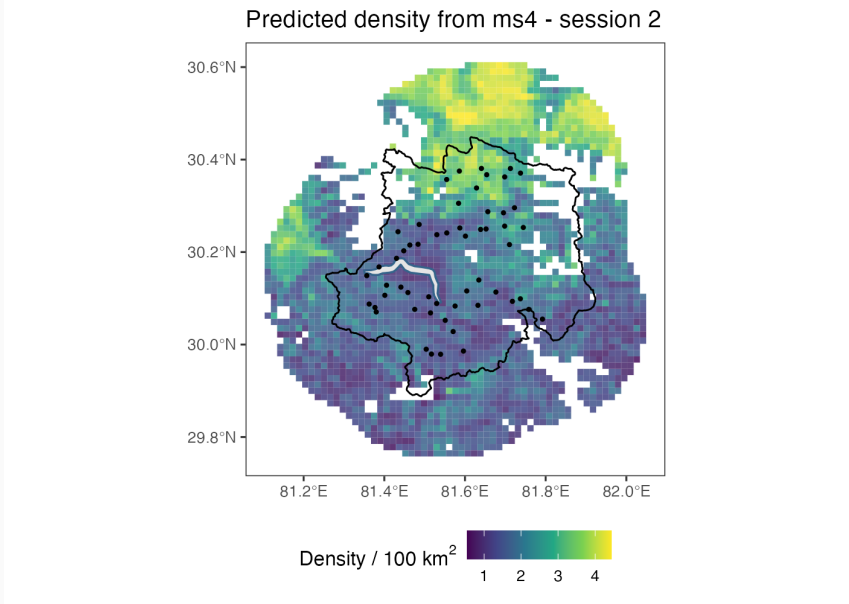


Predicted density maps

Multisession session 1



Multisession session 2



Workflow recap

1. Long surveys require thinking about closure.
2. Shortened surveys and multisession analyses change both captures and effort.
3. Habitat constraints change masks for fitting and reporting.
4. Barriers change movement distances, not the capture-history object.
5. Sex-specific models require careful handling of unknown sex and AIC comparisons.
6. Extrapolation to a new mask needs a `userdist` that matches.